

How to Install and Use Docker on Ubuntu

In the last few years, server virtualisation has become very popular. We cannot imagine cloud computing without virtualisation. But the explosive growth in computing demands more efficient virtualisation solutions. This is where containers come into play. Docker is one of the most popular container solution available today. In this article we will do hands-on with Docker.



Containers are lightweight virtualisation solutions. They provide OS-level virtualisation without any special support from the underlying hardware. Namespaces and control groups form the backbone of the container in the GNU/Linux kernel. Container solutions are built on top of these features. Namespaces provide isolation for processes, the network, mount points and so on, while control groups limit access to available hardware resources.

Sometimes, newbies get confused and think that server virtualisation and containerisation are the same thing. But, in fact, they are significantly different from each other. In server virtualisation, the OS is not shared; each VM instance has its own OS, whereas containers share the underlying OS. This approach has some advantages as well as disadvantages. The advantage is that the VM provides better isolation and security but performance is compromised. Whereas a container compromises on isolation but delivers a performance that's as good as bare hardware.

Containers have been around since quite a long time. Their roots can be found in UNIX's chroot program. After this release, many UNIX flavours have implemented their own container versions, like BSD jails and Solaris zones. On the GNU/Linux platform, LXD, OpenVZ and LXC are alternatives to Docker. However, Docker is much more mature and provides many advanced functionalities. We will discuss a few of them in the later sections of this article.

Setting up the environment

In this section, let's discuss how to install Docker on an Ubuntu distribution, a task that is as simple as installing other software on GNU/Linux. To install Docker and its components, execute the following commands in a terminal:

```
$ sudo apt-get update
$ sudo apt-get install docker docker.io docker-compose
```

That's it! The installation can be done by executing just two commands.

Now, let us verify the installation by printing the Docker version. If everything is fine, then it should display the installed Docker version. In my case, it was 1.13.1, as shown below:

```
$ docker --version
Docker version 1.13.1, build 092cba3
```

Now that we are done with the installation, let us briefly discuss a few important Docker components.

Docker Engine: This is the core of Docker. It runs as a daemon process and serves requests made by the client. It is responsible for creating and managing containers.

Docker Hub: This is the online public repository where Docker images are published. You can download as well as upload your custom images to this repository.